Reasoning-Aware Pruning of Large Language Models

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Introduction

- LLMs are generally over parameterized and pruning them removes unimportant weights to reduce inference cost.
- Standard pruning methods like Wanda preserve general language quality but dont consider reasoning.
- Our goal is to prune while considering reasoning capabilities.

Sparsity	GSM8K	ARC	PPL
Baseline	14.0%	33.0%	10.20
20%	12.0%	32.5%	10.67
40%	8.5%	29.0%	15.70
60%	6.5%	21.0%	209.96

Unpruned vs Wanda Pruning Results on Llama-3.2-1B

How Wanda Pruning works

- **Magnitude Pruning Flaw:** Keeps large weights, prunes small ones. But a large weight connected to an inactive input contributes nothing. A small weight on a highly active input matters a lot.
- Wanda Score: $\mathbf{score}(w_{ij}) = |w_{ij}| \times \|x_j\|_2$
- 3 steps:
  1. Run 32 batches of WikiText through model. Hooks collect input activations at every linear layer.
  2. For every weight: score = |weight| × activation norm. Both must be large to matter.
  3. Zero out bottom k% of scores. No retraining (one shot).

	Weight	Activation Norms	Wanda Score	Decision
W_A	0.8	0.1 (dead)	0.08	PRUNED
W_B	0.02	0.9 (active)	0.81	KEPT

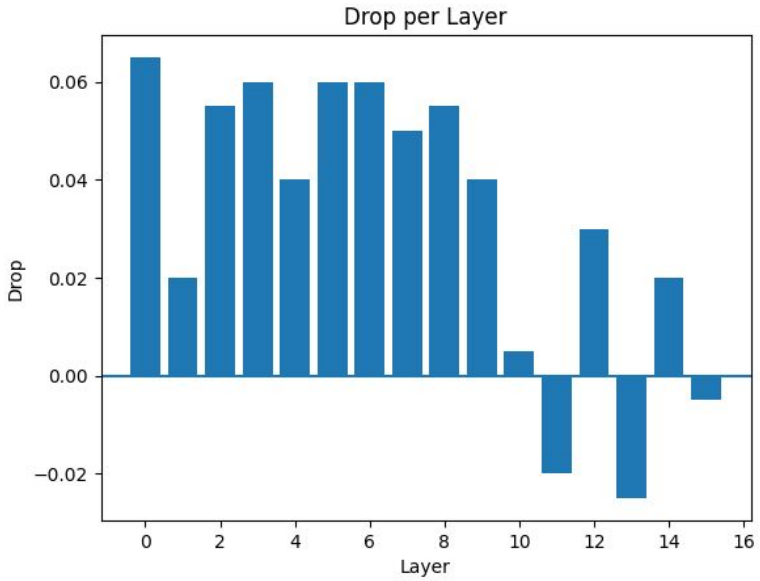
Wanda Pruning Example

Measuring Reasoning Sensitivity

Key Question: Which specific layers are responsible for reasoning?

For each layer i (16 total in LLaMA-3.2-1B):

- Attach hook → scale layer output by 0.5 during inference.
- Run 200 GSM8K questions with that layer weakened.
- Sensitivity = baseline accuracy − perturbed accuracy.
- Remove hook → layer fully restored → move to next layer.



Accuracy Drop per Layer

Our Proposed Pruning Metric

- final_score = α × wanda_normalized + (1 − α) × sensitivity_normalized
- **α** controls the tradeoff:

  α=1.0 → pure Wanda (baseline)

  α=0.5 → equal weighting

  α=0.0 → pure sensitivity (destroys model)

	Weight A	Weight B
Layer	Layer 0 (sensitive)	Layer 10 (not sensitive)
Wanda	0.2 (low)	0.3 (higher)
Sensitivity	1.0	0.08
Composite	0.60 → **KEPT**	0.19 → **PRUNED**

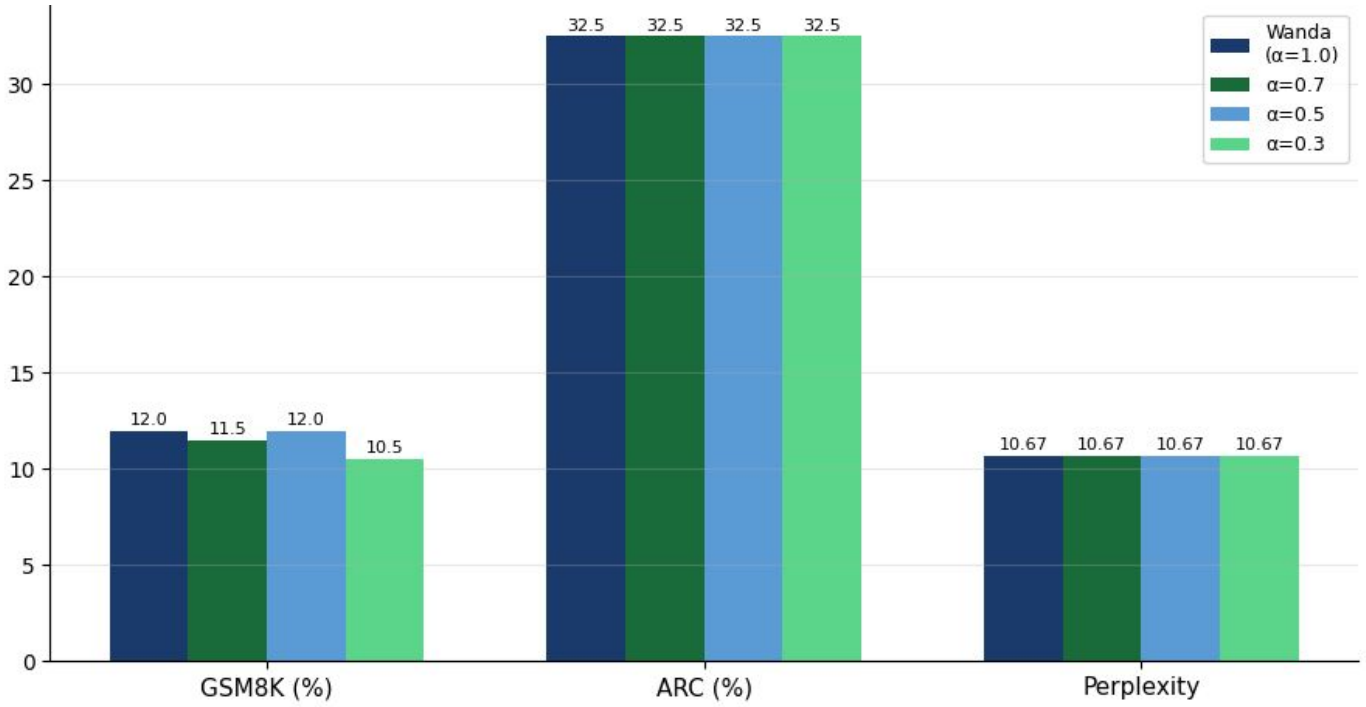
Example of Our Pruning Metric

Wanda vs Our Method

	Wanda (Baseline)	Our Method
Importance Signal	Weight magnitude × activation norm	Wanda score + reasoning sensitivity
Reasoning Awareness	None. It is blind to reasoning capability	Explicit via layerwise perturbation
Reasoning critical layers	Treated same as all others	Protected via sensitivity score
Calibration Cost	32 forward passes	32 + 16*200 GSM8K inferences

Both methods are one-shot. Our method adds reasoning awareness at the cost of 16 additional GSM8K evaluation passes.

Results



Pruning Comparison at 20% Sparsity

Nearly identical results to Wanda Pruning. **WHY?**

Potential Reasons

- Sparsity was too low to differentiate.
- 1B model has weak reasoning to begin with.
- Identical perplexity means nearly the same pruning masks.

How to overcome these Challenges?

Future Work

- Evaluate at higher sparsity levels (40%, 60%) to identify the crossover point where reasoning degrades disproportionately faster than perplexity.
- Scale experiments to larger models (3B+ parameters) with stronger baseline reasoning to better validate the sensitivity metric.
- Experiment with different perturbation scales (0.3 or 0.7) during sensitivity measurement to determine how it might affect which layers are defined as reasoning critical.

Member Contributions

- **Jai**: The layerwise perturbation experiment, sensitivity metric computation, and composite score formulation.
- **Shivang**: Wanda pruning baseline implementation and integration of sensitivity scores into the pruning pipeline.
- **Arnav**: Reasoning benchmark selection, Experimental evaluation framework, baseline comparisons, results analysis, and visualizations.